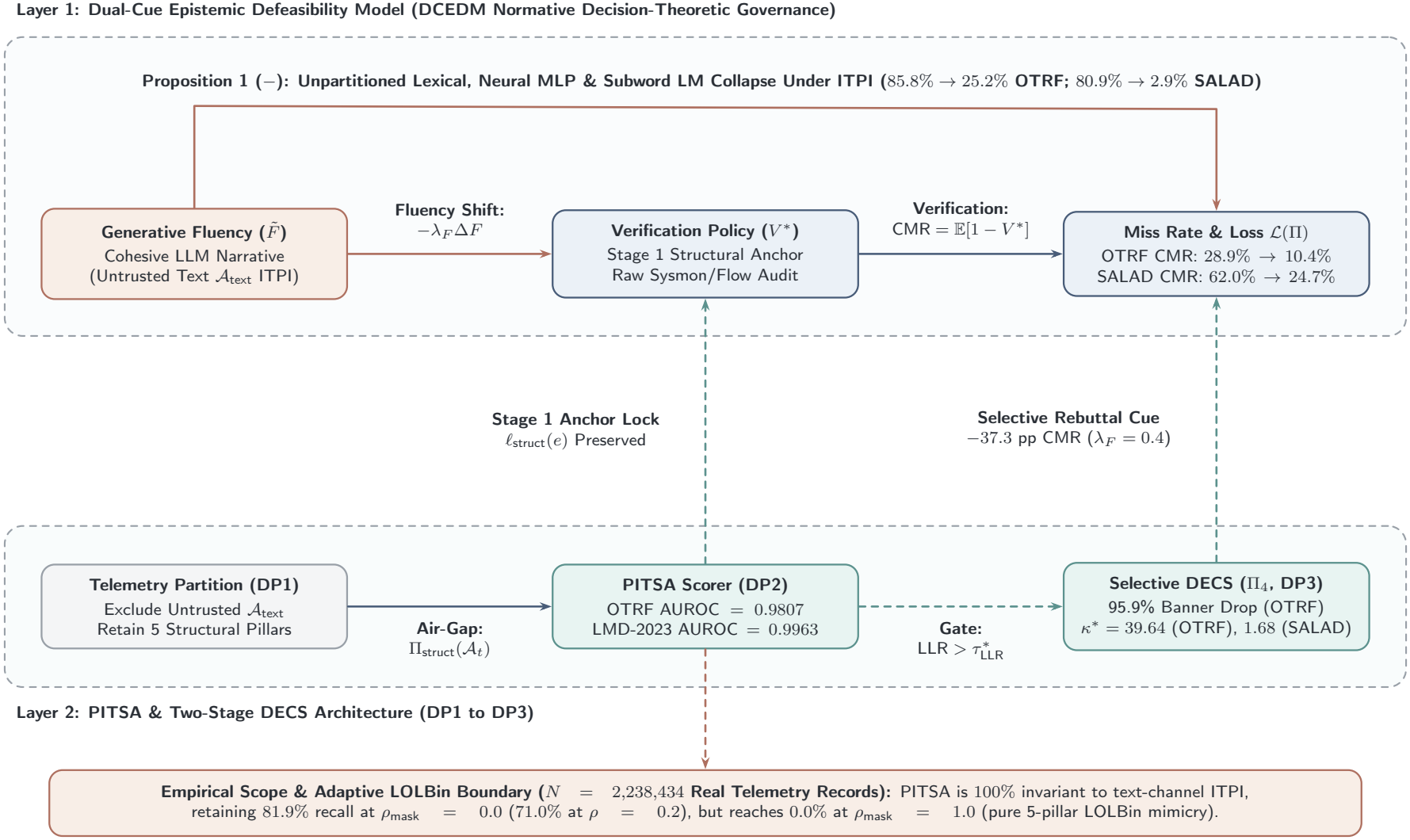

Layer 1: Dual-Cue Epistemic Defeasibility Model (DCEDM Normative Decision-Theoretic Governance)
Proposition 1 (−): Unpartitioned Lexical, Neural MLP & Subword LM Collapse Under ITPI (85.8% → 25.2% OTRF; 80.9% → 2.9% SALAD)
Generative Fluency ($\tilde{F}$)
Cohesive LLM Narrative
(Untrusted Text $\mathcal{A}_{\text{text}}$ ITPI)
Fluency Shift:
$-\lambda_F \Delta F$
Verification Policy (V^*)
Stage 1 Structural Anchor
Raw Sysmon/Flow Audit
Verification:
CMR $= \mathbb{E}[1 - V^*]$
Miss Rate & Loss $\mathcal{L}(\Pi)$
OTRF CMR: 28.9% → 10.4%
SALAD CMR: 62.0% → 24.7%
Stage 1 Anchor Lock
$\ell_{\text{struct}}(e)$ Preserved
Selective Rebuttal Cue
-37.3 pp CMR ($\lambda_F = 0.4$)
Telemetry Partition (DP1)
Exclude Untrusted $\mathcal{A}_{\text{text}}$
Retain 5 Structural Pillars
Air-Gap:
$\Pi_{\text{struct}}(\mathcal{A}_t)$
PITSA Scorer (DP2)
OTRF AUROC $= 0.9807$
LMD-2023 AUROC $= 0.9963$
Gate:
LLR $> \tau^*_{\text{LLR}}$
Selective DECS (Π_4, DP3)
95.9% Banner Drop (OTRF)
$\kappa^* = 39.64$ (OTRF), 1.68 (SALAD)
Layer 2: PITSA & Two-Stage DECS Architecture (DP1 to DP3)
Empirical Scope & Adaptive LOLBin Boundary ($N = 2{,}238{,}434$ Real Telemetry Records): PITSA is 100% invariant to text-channel ITPI, retaining 81.9% recall at $\rho_{\text{mask}} = 0.0$ (71.0% at $\rho = 0.2$), but reaches 0.0% at $\rho_{\text{mask}} = 1.0$ (pure 5-pillar LOLBin mimicry).